

CSA 1718 – Artificial Intelligence

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Prolog Program for Backward Chaining

Aim

To write and implement a Prolog program to demonstrate the concept of backward chaining.

Objective

- To understand backward chaining.
- To start with a goal and work backward toward supporting facts.
- To implement logical reasoning using Prolog.

Theory

Backward chaining starts with a goal and tries to find facts or rules that can prove the goal.

For example:

Goal: Ground is wet.

↓

Rule: If it rains, ground is wet.

↓

Need to prove: It rains.

↓

Fact: It rains.

Therefore, the goal is proved.

Algorithm

1. Define the required basic fact.
2. Define a rule that can prove the goal.
3. Start with the goal.
4. Prolog searches for a rule that can satisfy the goal.
5. It then tries to prove the conditions of that rule.

6. If the conditions are true, the goal succeeds.

Program

rain.

wet_ground :-

rain.

umbrella :-

rain.

Query

wet_ground.

Output

true.

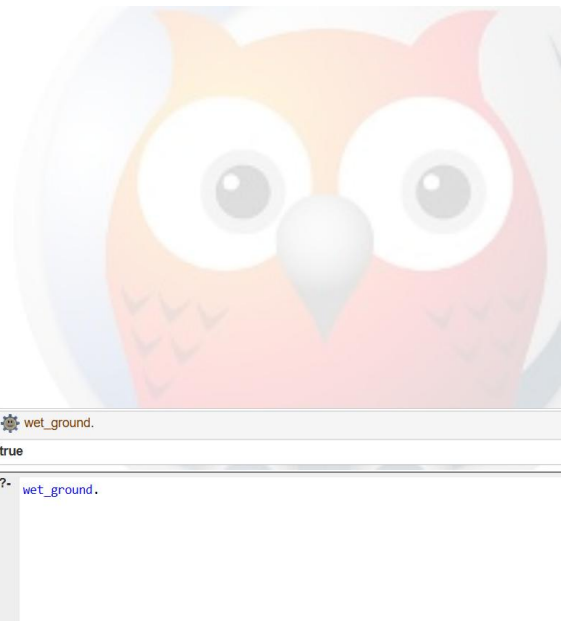
Another query:

umbrella.

Output:

true.

```
1 rain.  
2  
3 wet_ground :-  
4   rain.  
5  
6 umbrella :-  
7   rain.
```



Result

Thus, the concept of backward chaining was successfully demonstrated using Prolog.

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